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External laser enabled co-packaged optics architectures

Abstract

A sourceless co-packaged optical-electrical chip can include a plurality of different optical transceivers, each of which can transmit to an external destination or internal components. Each of the transceivers can be configured for a different modulation format, such as different pulse amplitude, phase shift key, and quadrature amplitude modulation formats. Different light sources provide light for processing by the transceivers, where the light source and transceivers can be configured for different applications (e.g., different distances) and data rates. An optical coupler can combine the light for the different transceivers for input into the sourceless co-packaged optical-electrical chip via a polarization maintaining media (e.g., polarization maintaining few mode fiber and polarization maintaining single mode fiber), where another coupler operates in splitting mode to separate the different channels of light for the different transceivers according to different co-packaged configurations.

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Background/Summary

PRIORITY (1) This application is a continuation of U.S. patent application Ser. No. 17/470,650, filed Sep. 9, 2021, which is a continuation of U.S. patent application Ser. No. 16/990,348 filed Aug. 11, 2020, which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

(1) The present disclosure generally relates to optical and electrical devices and more particularly to integration of sourceless co-packaged silicon switch and photonic circuits.

BACKGROUND

(2) The growth of network traffic of data centers has driven a demand for higher capacity and lower power consumption. One approach is moving the optical components of chips (e.g., optical transceivers) closer to the silicon switch and integrating some of the IC components traditionally on the host board. In some approaches, the silicon switch and optical/electrical components can be co-packaged on a single chip. While closely integrating the electrical and optical components can, in theory, enable higher capacities (e.g., smaller footprint) with lower power consumption and increased data speeds, these co-packaged approaches are difficult to design and often cannot be changed or reconfigured down the road if one or more devices fail.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following description includes discussion of figures having illustrations given by way of example of implementations of embodiments of the disclosure. The drawings should be understood by way of example, and not by way of limitation. As used herein, references to one or more “embodiments” are to be understood as describing a particular feature, structure, or characteristic included in at least one implementation of the inventive subject matter. Thus, phrases such as “in one embodiment” or “in an alternate embodiment” appearing herein describe various embodiments and implementations of the inventive subject matter, and do not necessarily all refer to the same embodiment. However, they are also not necessarily mutually exclusive. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure (“FIG.”) number in which that element or act is first introduced.

(2) FIG. 1 is a block diagram illustrating a sourceless optical transceiver for transmitting and receiving optical signal, according to some example embodiments

(3) FIG. 2 is an illustration of a sourceless opto-electrical device (e.g., optical transceiver, sourceless co-packaged optical Silicon switch) including one or more optical devices, according to some example embodiments.

(4) FIG. 3 shows an external laser-enabled sourceless co-packaged optics architecture, according to some example embodiments.

(5) FIG. 4 shows an example mode coupler architecture, according to some example embodiments

(6) FIG. 5 shows a sourceless optical subsystem architecture implementing quantum dot lasers as external light sources, according to some example embodiments.

(7) FIG. 6 shows an optical sourceless co-packaged architecture, according to some example embodiments.

(8) FIG. 7 shows a flow diagram of a method for routing light using a laserless co-packaged architecture, according to some example embodiments.

(9) Descriptions of certain details and implementations follow, including a description of the figures, which may depict some or all of the embodiments described below, as well as discussing other potential embodiments or implementations of the inventive concepts presented herein. An

overview of embodiments of the disclosure is provided below, followed by a more detailed description with reference to the drawings.

DETAILED DESCRIPTION

(10) In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide an understanding of various embodiments of the inventive subject matter. It will be evident, however, to those skilled in the art, that embodiments of the inventive subject matter may be practiced without these specific details. In general, well-known instruction instances, structures, and techniques are not necessarily shown in detail.

(11) As discussed, co-packaged electrical-optical designs for optical communication modules (e.g., transceiver chips, switch chips) can reduce the power consumption, but it can be difficult to implement the co-packaged approaches in a way that provides reconfigurability for light sources and component failures. To this end, an optical sourceless co-packaged architecture can be implemented to offer the benefit of co-packaged optic architecture low power consumption while enabling reconfigurability of the architecture for different modulation formats and optical sources. The optical sourceless co-packaged architecture can receive light from one or more light banks, or laser sub-assemblies that are external to the optical sourceless co-packaged architecture and can be changed as required per implementation. In some example embodiments, the optical sourceless co-packaged architecture comprises a plurality of optical transmitters that can transmit in different modulation formats, such as m-ary pulse amplitude modulation (M-PAM), or m-ary quadrature amplitude modulation (M-QAM). The different channels or beams of light from the different light sources that are destined to different transceiver modules can be combined and separated using one or more spatial mode couplers that convert each channel of light into a different higher order optical mode and couple the channels in the different higher order modes on a fiber that transmits multiple optical modes (e.g., few mode fiber). The optical sourceless co-packaged architecture is coupled to the fiber and receives the different channels and separates them based on their different spatial modes using another mode coupler that operates as a mode-based splitter that also converts each channel of light into a lower order mode (e.g., fundamental mode, Gaussian mode). The separated channels are then input to different transmitter modules for modulation and transfer to various destinations (e.g., remote receivers, internal receivers), as discussed in further detail below.

(12) FIG. 1 is a block diagram illustrating a sourceless optical transceiver **100**, according to some example embodiments. The optical transceiver **100** is an example of a co-packaged chip architecture that can include one or more transceiver modules in a photonic integrated circuit (PIC) that modulate light from an external light source **117** (e.g., laser bank, quantum dot lasers, etc.) that is coupled via a fiber (e.g., few mode fiber).

(13) As illustrated, the sourceless optical transceiver **100** is implemented to interface electrical data from electrical devices, such as electrical hardware device **150**; converts the electrical data into optical data; and sends and receives the optical data with one or more optical devices, such as optical device **175** (e.g., an external transmitter **177** of a remote optical device, switch, server, etc.). For explanatory purposes, in the following description, the electrical hardware device **150** is a host board that “hosts” the optical transceiver **100** as an optical engine that sends and receives data to an optical switch network, where, for example, optical device **175** can be other components of an optical switch network (e.g., IP/L2 switch, OTN (L1) switch). However, it is appreciated that the system can be implemented to interface other types of electrical devices and optical devices. For instance, optical transceiver **100** can be implemented as a single chip on a hybrid Ethernet switch co-package that uses an optical network (e.g., waveguides, fibers) as an optical bus to interconnect on-board electrical chips that process the data after it is converted from light into binary electrical data, according to some example embodiments.

(14) In some example embodiments, the hardware device **150** includes an electrical interface for communicating with an electrical interface of the optical transceiver **100**. The optical transceiver **100** may be a removable front end module that may be physically received by and removed from

hardware device **150** operating as a back end module within a communication system or device. The optical transceiver **100** and the hardware device **150**, for example, can be components of an optical communication device (e.g., a network device) such as a single channel intensity modulator/direct detection (IM/DD) transceiver, single channel coherent transceiver, wavelength-division multiplexing (WDM) based multi-channel IM/DD transceiver, or WDM based multi-channel coherent transceiver. M-ary PAM formats can be used in the IM/DD transceiver, while M-ary phase-shift keying (M-PSK) or M-ary quadrature amplitude modulation (M-QAM) formats can be used in the coherent transceiver.

(15) A data transmitter **105** of the optical transceiver **100** can receive the electrical signals from hardware device **150**, which are then converted into optical signals via one or more transmitter modules in the PIC **110** (e.g., an optical modulator, waveguides, a modulator driver). The PIC **110** can then output the optical signals via optical links, such as fiber or waveguides that interface with the PIC **110**, to other components. The output light data can then be processed by other components (e.g., switches, endpoint servers, other embedded chips of a single embedded system), via a network such as a wide area network (WAN), optical switch network, optical waveguide network in an embedded system, and others.

(16) In receiver mode, the optical transceiver **100** can receive high data rate optical signals via one or more optical links from optical device **175**. The optical signals are converted by the PIC **110** from light into electrical signals for further processing by data receiver **115**, such as demodulating the data into a lower data rate for output to other devices, such as the electrical hardware device **150**. In some example embodiments, the data transmitter **105** and data receiver **115** are formed as a single electrical circuit structure, such as ASIC **103**. The ASIC **103** can further include other circuit structures, including, for example, a switch ASIC or one or more digital signal processing (DSP) portions to provide further data processing functionality.

(17) FIG. 2 is an illustration of a sourceless opto-electrical device **200** (e.g., optical transceiver **100**, sourceless co-packaged optical switch **605**) including one or more optical devices, according to some example embodiments. In this embodiment, the sourceless opto-electrical device **200** is shown to include printed circuit board (PCB) substrate **205**, organic substrate **260**, and a co-packaged architecture **222**, which includes an application-specific integrated circuit (ASIC) **215** (e.g., data transmitter **105**, data receiver **115**, packet-digital switches **340A/340B**, ASIC **610**) and PIC **220** (e.g., transmitter components, such as co-packaged modules **330**, **335** of FIG. 3 discussed below).

(18) In some example embodiments, the PIC **220** includes silicon on insulator (SOI) or silicon based (e.g., silicon nitride (SiN)) devices, or may comprise devices formed from both silicon and a non-silicon material. Said non-silicon material (alternatively referred to as “heterogeneous material”) may comprise one of III-V material, magneto-optic material, or crystal substrate material. III-V semiconductors have elements that are found in group III and group V of the periodic table (e.g., Indium Gallium Arsenide Phosphide (InGaAsP), Gallium Indium Arsenide Nitride (GaInAsN)). The carrier dispersion effects of III-V-based materials may be significantly higher than in silicon-based materials, as electron speed in III-V semiconductors is much faster than that in silicon. In addition, III-V materials have a direct bandgap, which enables efficient creation of light from electrical pumping. Thus, III-V semiconductor materials enable photonic operations with an increased efficiency over silicon for both generating light and modulating the refractive index of light. Thus, III-V semiconductor materials enable photonic operation with an increased efficiency at generating light from electricity and converting light back into electricity.

(19) The low optical loss and high quality oxides of silicon are thus combined with the electro-optic efficiency of III-V semiconductors in the heterogeneous optical devices described below; in embodiments of the disclosure, said heterogeneous devices utilize low loss heterogeneous optical waveguide transitions between the devices' heterogeneous and silicon-only waveguides.

(20) Magneto-optic (MO) materials allow heterogeneous PICs to operate based on the MO effect.

Such devices may utilize the Faraday Effect, in which the magnetic field associated with an electrical signal modulates an optical beam, offering high bandwidth modulation, and rotates the electric field of the optical mode, enabling optical isolators. Said MO materials may comprise, for example, materials such as iron, cobalt, or yttrium iron garnet (YIG). Further, in some example embodiments, crystal substrate materials provide heterogeneous PICs with a high electro-mechanical coupling, linear electro-optic coefficient, low transmission loss, and stable physical and chemical properties. Said crystal substrate materials may comprise, for example, lithium niobate (LiNbO₃) or lithium tantalate (LiTaO₃).

(21) In the example illustrated, the PIC **220** exchanges light with an external light source **225** (e.g., light source **117**, light source subassembly **305**) via fiber **221** (e.g., few mode fiber (FMF), multicore fiber). The fiber **221** can couple with the PIC **220** using a prism, grating, or lens, according to some example embodiments. In other example embodiments, multiple fibers are implemented to receive light at different light sources, as discussed in further detail below with reference to FIG. 6

(22) In some example embodiments, the optical devices of PIC **220** are controlled, at least in part, by control circuitry included in ASIC **215**. Both ASIC **215** and PIC **220** are shown to be disposed on copper pillars **214**, which are used for communicatively coupling the PICs via organic substrate **260**. PCB substrate **205** is coupled to organic substrate **260** via ball grid array (BGA) interconnect **216** and may be used to interconnect the organic substrate **260** (and thus, ASIC **215** and PIC **220**) to other components of the sourceless opto-electrical device **200** not shown (e.g., interconnection modules, power supplies, etc.).

(23) FIG. 3 shows a sourceless optical transmitter architecture **300**, according to some example embodiments. A light source subassembly **305** (e.g., laser bank, optical box with a set of laser diodes), with multiple light sources functions as an external source of light to be input into the plurality of sourceless co-packaged optics modules, including PAM module **330** and coherent module **335**. In some example embodiments, one or more of the light sources in the light source subassembly **305** can be a different type of light source (e.g., distributed feedback laser, quantum dot laser, an external cavity laser). In some example embodiments, each of the included light sources have different optical characteristics, such as linewidth, optical power, or relative intensity noise.

(24) The sourceless optical transceiver **100** is laserless or sourceless that can operate using light provided from an external source. That is, for example, the sourceless optical transceiver **100** would still be laserless with respect to the PAM module **330** and the coherent module **335** even where the co-packaged chip (e.g., sourceless optical transceiver **100**, sourceless co-packaged optical switch **605**) includes an optical source, so long as one or more of the CPO modules in the chip receives light from an optical source (e.g., via fiber), according to some example embodiments. In the illustrated embodiments, the light source subassembly **305** comprises a plurality of lasers, including a first set of lasers **0**, **1**, **2**, **3**, **X** and a second set of lasers **4**, **5**, **6**, **7**, **Y**.

(25) The first set of lasers **0**, **1**, **2**, **3**, and **X** can be configured to generate multiple wavelengths of light, λ_1 - λ_4 , at the same or different powers, for use by a transmitter module, such as the PAM module **330**, according to some example embodiments. In particular, for example, laser **0** generates λ_1 light at 1 decibel-milliwatt (dBm) output power and 1 megahertz (MHz) linewidth, laser **1** generates λ_3 light at 4 dBm output power and 1 MHz linewidth, laser **2** generates λ_2 light at 7 dBm output power and 1 MHz linewidth, laser **3** generates λ_4 light at 10 dBm output power and 1 MHz linewidth, and laser **X**, which is a tunable laser that generates light having changeable characteristics. In some example embodiments, laser **X** is a redundancy light source that can be configured to generate light having any power to 13 dBm, and a tunable frequency range that can be configured for any frequency between λ_1 to λ_4 . Laser **X** can be configured to generate a replacement channel light beam to replace any of the lasers **0**, **1**, **2**, **3** should they malfunction. For example, if light from laser **1** is not being received by the PAM module **330**, adaptive selection

controller **360** (e.g., a user device, desktop PC) can use a data interface to receive data indicating one of the lasers is malfunctioning, and use the data interface to tune laser X to generate λ_3 light at 4 dBm power to replace the malfunctioning laser **1**. In some example embodiments, multiple lasers generate light at the same time for a given transmitter. For example, lasers **0-3** can simultaneously generate light which is then combined and input into the PAM module **330** as discussed here (e.g., spatial mode coupling for combining and separating the multi-wavelength light destined for the PAM module **330** from light destined to other transmitter modules).

(26) The second set of lasers **4, 5, 6, 7**, and Y can be configured to generate multiple wavelengths of light, λ_1 - λ_4 , at the same or different powers and linewidth, for use by another transceiver module, such as the coherent module **335**, according to different example embodiments. In particular, for example, laser **4** generates λ_4 light at 15 dBm output power and 10 kHz linewidth, laser **5** generates λ_3 light at 12 dBm output power and 100 kHz linewidth, laser **6** generates λ_2 light at 9 dBm output power and 200 kHz linewidth, laser **7** generates λ_1 light at 6 dBm output power and 500 kHz linewidth, and laser Y, which is a tunable laser that can generate different types of light. In some example embodiments, laser Y is a redundancy light source for the second set of lasers that can be configured to generate light having any power up to 18 dBm, and a tunable frequency range that can be configured for any frequency between λ_1 to λ_4 . Laser Y can be configured to function as a replacement channel light source to replace any of the lasers **4, 5, 6, 7** should they malfunction. For example, if light from laser **7** is not being received by the coherent module **335**, then the adaptive selection controller **360** (e.g., client device) can use a data interface (e.g., serial interface) to receive data indicating one of the lasers is malfunctioning, and use the data interface to tune laser Y to generate λ_1 light at 6 dBm power to replace the malfunctioning laser **7**.

(27) The PAM CPO module **330** can receive the light generated by one or more of the first set of lasers in the light source subassembly **305** and input the light into one or more PAM4 transceiver **350** using MUX **345A** (e.g., a polarization maintaining multiplexer in demultiplexer/splitting mode). The transceivers **350** are PAM4 transceivers of different types including a PAM4-SR transceiver ("TRxO") transmitting/receiving PAM4 data over 100 meters using single or multiple fibers, PAM4-FR transceiver ("TRx1") transmitting/receiving PAM4 data over 2 kilometers using single or multiple fibers, PAM4-DR transceivers ("TRx2") transmitting/receiving PAM4 data over 500 meters using single or multiple fibers, and PAM4-LR transceiver ("TRx3") transmitting/receiving PAM4 data over 10 kilometers. Although in the example of FIG. **3** (and FIG. **5**), the PAM CPO module **330** comprises four transceivers, it is appreciated that the PAM CPO module **330** can include a different number of transceivers (e.g., 1 PAM transceiver, 20 PAM transceivers), or transmitters without receivers, and so on, according to some example embodiments.

(28) The coherent CPO module **335** is a coherent module that can receive light generated by one or more of the second set of lasers in the light source subassembly **305** and input the light into one or more coherent transmitters **355** using MUX **345B** (in demux mode). For example, the coherent module **355** is configured to a coherent detector receiver that mix the local oscillator light with the optical signals to demodulate the received coherent light beam data. The transmitters **355** are coherent transmitters of different types including a QPSK transmitter ("TRxO"), an 8 QAM transmitter ("TRx1"), a 16 QAM transmitter ("TRx2"), and a 64 QAM transmitter ("TRx3"). Although in the example of FIG. **3** (and FIG. **5**), the coherent CPO module **335** comprises four transceivers, it is appreciated that the coherent CPO module **335** can include a different number of transceivers (e.g., 1 coherent transceiver, 20 coherent transceivers), or coherent transmitters without receivers, and so on, according to some example embodiments.

(29) Further, although only two CPO modules are illustrated in FIG. **3**, it is appreciated that additional modules can be implemented, such as a WDM module in which all the transmitters receive the low power of light at different wavelengths to transmit WDM a short distance (e.g., 500 m) over a single fiber, and a second WDM module in which all the transmitters in the second

WDM module receive high power light at different wavelengths to transmit WDM data for longer distances (e.g., 2 km) over a single mode fiber. Further, in the example illustrated, each of the modules includes electronic control circuitry, such as a packet-digital circuit **340A** and **340B**, to communicate with different transceivers, while in other example embodiments, the electronic control circuit is located in another circuit structure, such as ASIC **610** in FIG. **610**.

(30) The light from the different sets of lasers in the light source subassembly **305** are directed towards specific CPO modules using few-mode couplers, including spatial mode couplers **315** and **325**, and a polarization maintaining few mode fiber **320** (PM-FMF), which support different spatial modes of light. Generally, few mode components (e.g., fibers, couplers) transmit multiple mode channels (multiple wavelength channels can be used to transmit light in each mode) in a single component (e.g., fiber, coupler), where each mode channel takes an independent spatial pathway (e.g., different modes). The channels may be multiplexed together using a spatial mode multiplexer and transmitted across a multi-spatial mode fiber (e.g., a few mode fiber, which is physically larger than single mode fiber and configured to support several transverse spatial modes on a single medium, as opposed to a multicore fiber which transmits different spatial channels on different cores).

(31) The spatially multiplexed channels may be received using a spatial mode demultiplexer, which separates the channels by their different spatial modes for further routing (e.g., to different receivers for each channel, etc.). As an example, and with reference to FIG. **3**, light from one or more lasers in the first set of lasers **307** is coupled via a polarization maintaining single mode coupler **310A** to single mode fiber **313** (a polarization maintaining single mode fiber), which is further coupled to the few-mode coupler **315**. Similarly, light from one or more lasers in the second set of lasers **311** is coupled via a polarization maintaining single mode coupler **310B** along another single mode fiber **317** to the few mode coupler **315**. The mode coupler **315** then converts the received light into different modes, combines them, and outputs the mode light onto the polarization maintaining FMF **320** (e.g., a few-mode fiber that supports several linearly polarized (LP) spatial modes, such as LP.sub.01, LP.sub.11, LP.sub.21, LP.sub.02) and different orientations, such as LP.sub.11 and LP.sub.21, discussed below). In particular, for example, the mode coupler **315** converts the light from the fiber **313** to a first mode (e.g., mode-1, LP.sub.11), and light from the other fiber **317** to a second different mode (e.g., mode-2, LP.sub.21), which are combined and output onto FMF **320**.

(32) The FMF **320** is then coupled to another mode coupler **325**, which is configured to separate or demultiplex the received channels (e.g., mode-1 and mode-2) onto different paths. For example, the mode coupler **325** converts the first mode of light (mode-1, from one or more of the first set of lasers **307**) into a fundamental mode (e.g., Gaussian mode, LP.sub.01), and inputs the light into the PAM module based co-packaged optics **330** for processing (e.g., via one or more of the PAM4 transceivers **350**). Further, the mode coupler **325** converts the second mode of light (mode-2, from the second set of lasers **311**) into a Gaussian mode on another fiber, which then inputs the light into the coherent CPO module **335** for processing (e.g., via one or more of the transmitters **355** for different types of coherent modulation-based communication). Further, a portion of the light input into the coherent CPO module **335** may remain unmodulated and be transmitted to the receiver at the destination for coherent signal processing. In some example embodiments, the mode coupler **325** is an embedded spatial mode coupler that is integrated into a sourceless CPO chip architecture, such as sourceless optical transceiver **100** or sourceless co-packaged optical switch **605**.

(33) FIG. **4** shows an example mode coupler architecture **400**, according to some example embodiments. In the illustrated example, the mode coupler architecture **400** includes two couplers **405** and **410** (e.g., 2×2 couplers), which can generate and separate higher spatial modes. In particular, the mode coupler **315** (FIG. **3**) can implement mode coupler architecture **400** to generate higher order spatial modes (e.g., LP11, LP21) in a multiplexer or combining mode (input light on the single mode ports **403** on the left side of mode coupler architecture **400**; output light on the port

407, on the right side of the mode coupler architecture **400**). Further, the mode coupler **325** (FIG. 3) can be implemented as a reverse configuration of mode coupler architecture **400**, in which light is received on the port **407** (from FMF **320**, FIG. 3) and separates higher order spatial modes into Gaussian modes in demultiplexer or splitter mode (from right to left). Although in the example of FIG. 4 a two coupler architecture is implemented for the mode couplers **315**, **325** it is appreciated that other configurations with additional components (e.g., splitters, gratings) can be implemented to provide spatial mode coupling combining and separating functionality to provide light to the sourceless optical transceiver **100**, according to different example embodiments.

(34) As an illustrative example, the light from the fiber **313** is split onto two single mode fibers, SMF-1 and SMF-2, each of which carries the light in the Gaussian mode LP01, which are input into coupler **405** to generate higher order light LP11, and LP01 (which can be filtered or discarded). The light from the coupler **405** is then input into one of the coupler ports of coupler **410** (via FMF connecting the couplers **405** and **410**), where the other coupler port of **410** receives light from the fiber **317** for coupling with the light from the coupler **405**. That is, the light from fiber **317** is input directly into coupler **410** using another single mode fiber, SMF-3, in its Gaussian spatial mode LP01.

(35) The coupler **410** then couples the light from coupler **405** and the light from fiber **317** to generate high-order mode light **415**, which can propagate over few mode fiber (e.g., FMF **320**, FIG. 3) to its destination, such as the mode coupler **325** on the sourceless optical transceiver **100**. The mode coupler **325** can have two other couplers in the reverse configuration illustrated in mode coupler architecture **400** to separate the light to the different modules. That is, for example and with reference to mode coupler architecture **400**, the mode coupler receives light on the port **407**, and converts one channel of light in higher mode LP21 into Gaussian mode LP01, which is then output onto SMF-3, which is then input into one of the transmitter modules (e.g., PAM module **330**). Further, the other higher mode of light LP11 (along with LP01) is output by the port of coupler **410** (on the left side of coupler **410**), which is then separated further by coupler **405** to generate two beams each in Gaussian mode onto SMF-1 and SMF-2, where one of the fibers can be used to input that channel of light into one or more other transmitter modules, such as coherent module **335**. As discussed, it is appreciated that different mode coupler configurations can be implemented to yield different configurations of outputs and powers, according to how many transmitter modules are to receive light from the external sources.

(36) FIG. 5 shows a sourceless optical subsystem architecture **500** implementing quantum dots as light sources, according to some example embodiments. In contrast to the sourceless optical transmitter architecture **300** in FIG. 3, the total number of lasers is reduced and the quantum dot lasers are implemented to generate multiple wavelengths of coherent light in a single beam, which then can be filtered to select the desired beam for output. The quantum dot lasers can be fabricated as monolithic light sources grown directly on silicon (e.g., InGaAs quantum dots). In each of the quantum dot lasers **505A** and **505B**, an active region between the p-cladding region and the n-cladding region comprises one or more quantum dots, each of which can generate light at a certain wavelength depending on the size and other characteristics of a given dot. Upon applying electricity to the active region, multi-beam coherent quantum laser light is generated for output, where power is proportional to the quantum dot density in the active region between the claddings and upon the applied electricity (e.g., current, voltage).

(37) As an example, the quantum dot laser **505A** can generate light that contains multiple beams of coherent light (e.g., λ_1 , λ_2 , λ_3 , λ_4) and one of the beams can be selected by filtering out the other beams (e.g., filtering out λ_2 - λ_4) to generate the desired output light from the quantum dot laser **505A** (e.g., λ_1). Similarly, laser **505B** can be a differently configured quantum dot laser (higher power) that generates multiple beams of coherent light at one or more wavelengths, which are similarly filtered to generate desired output light from the quantum dot laser **505B**. The light from each of the quantum dot lasers **505A** and **505B** can then be combined via the few mode fiber and

mode couplers, as discussed above.

(38) FIG. 6 shows an optical sourceless co-packaged architecture **600**, according to some example embodiments. In the example illustrated, the sourceless co-packaged optical switch **605** includes an electrical circuit, e.g., ASIC **610**, that provides electrical interconnections for optical components, including, for example a first set of optical transceivers **615** and a second set of transceivers **620**. The light processed by the optical sourceless co-packaged architecture **600** can be multidimensional and different types of transceivers and optical components can combine, separate, and process the light using different mechanisms (e.g., wavelength based coupling, spatial mode based coupling, and beam splitting), according to some example embodiments.

(39) The sourceless co-packaged optical switch **605** is laserless; light is input into the sourceless co-packaged optical switch **605** by external light sources, laser subassembly **635** and laser subassembly **640**. The laser subassembly **640** can include multiple lasers, which are input via a polarization maintaining single mode fiber **645** (e.g., carrying multiple wavelengths of light) that is coupled into the sourceless co-packaged optical switch **605** by the optical interface **655** (e.g., splitter, demultiplexer). The light at different wavelengths propagates in the fiber **645** and can be amplified by a fiber amplifier **650** (e.g., polarization maintaining-Erbium Doped Fiber Amplifier (PM-EDFA)) and then input into the chip by optical interface **655**, which splits and couples the light to the transceivers **620** via waveguides **625**.

(40) The laser subassembly **635** includes a set of light sources that generate light from the set of optical transceivers **615** for processing (e.g., transmitting and receiving via the ASIC **610**). In the example of FIG. 6, a single mode fiber **660** and few-mode fiber **665** each carry light that is input into the couplers, including splitter **672** and few-mode coupler **670** (e.g., spatial mode coupler). The two fibers **660** and **665** can be part of a ribbon cable that connects to connector **670** of the sourceless co-packaged optical switch **605**. The laser subassembly **635** can include two mode couplers that generate light for each of the fibers **660** and **665**, which are then separated or decoupled via the coupler **672** (via splitting) and coupler **670** (e.g., via spatial mode decoupling).

(41) FIG. 7 shows a flow diagram of a method **700** for processing light using a laserless co-packaged architecture, according to some example embodiments. At operation **705**, a first external light source (e.g., any of lasers **0**, **1**, **2**, **3**, or **X**) generates a first light beam and a second external light source (e.g., any of lasers **4**, **5**, **6**, **7**, or **Y**) generates a second light beam.

(42) At operation **710**, a coupler combines multidimensional light (e.g., light in different wavelengths, light in different spatial modes, light on different fiber cores). For example, at operation **710**, the mode coupler **315** converts the light received from laser **5** from its original base mode (Gaussian) to a higher order mode (LP11) and further converts the light from laser **7** from its base mode to another higher order mode (LP21), and combines the higher order modes to generate multimode light. As an additional example, at operation **710**, the laser subassembly **640** generates multi-wavelength light (WDM).

(43) At operation **715**, the multidimensional light is input into an optical sourceless co-packaged chip. For example, at operation **715**, the multimode light propagates on a few mode fiber (e.g., FMF **320**, FMF **665**) and is input to the chip via an optical connector (e.g., FMF coupler **670**, a lens, a grating). Additionally, at operation **715**, multiwavelength light on a single fiber is input into the optical sourceless co-packaged chip, according to some example embodiments.

(44) At operation **720**, the multidimensional light is separated. For example, at operation **720**, an embedded optical spatial mode decoupler (e.g., coupler **325**) receives the multimode light and separates the light onto different single mode channels, where the light is both converted from higher modes to the lower mode and separated by the mode decoupler. Additionally, at operation **720**, multiwavelength light propagating on single mode fiber **645** is input into the optical interface **655**, according to some example embodiments.

(45) At operation **725**, the first light set, and second light set are input into one or more CPO modules. For example, the first light set is input in its Gaussian mode into a PAM 4 CPO module,

such as PAM CPO module **330** and the second light set is input in its Gaussian mode into a coherent CPO, such as coherent CPO module **335**. In some example embodiments, the first light set and the second light set are input to the same type of CPO modules (e.g., both first and second light are input into direct detection modules, both first and second light are input into coherent modules). Additionally, and in accordance with some example embodiments, at operation **725**, the multiwavelength light is separated by the optical interface **655** and input into the transceivers **620**. (46) At operation **730**, the first and second light sets are modulated by their respective transmitters. For example, the PAM module **330** uses optical modulators in its SR transceiver (e.g., TRxO of PAM transceivers **350**, FIG. 3) to modulate PAM4 data onto the first light set. Further, the coherent module **335** uses optical modulators in its 16 QAM transceiver to modulate 16 QAM data onto the second light, and further input the second light set to the receiver for coherent demodulation and processing.

(47) At operation **735**, the modulated first and second lights are transmitted. For example, the PAM CPO module **330** transmits the PAM4 modulated light (e.g., to an external network or into an internal switch ASIC, such as ASIC **610**) and the coherent CPO module **335** transmits the 16 QAM modulated light (e.g., to an external network or device or into an internal ASIC, such as ASIC **610**, for processing).

(48) The following are example embodiments:

(49) Example 1. A method comprising: receiving, by a co-packaged optical device, multimode light comprising a plurality of beams in different modes including a first mode and a second mode, the multimode light received from a few mode fiber coupled to the co-packaged optical device, the co-packaged optical device comprising a plurality of co-packaged optics modules configured to transmit and receive light in different modulation formats, the plurality of co-packaged optics modules including at least one or more direct detection transceivers or coherent transceivers; separating, using a coupler of the co-packaged optical device, a first beam of light from the first mode and a second beam of light from the second mode; modulating the first beam of light using one of the co-packaged optics modules and modulating the second beam of light using another of the co-packaged optics modules; and transmitting the first beam of light using the one of the co-packaged optics modules and transmitting the second beam of light using the another of the co-packaged optics modules.

(50) Example 2. The method of example 1, wherein the one of the co-packaged optics modules comprises direct detection transceivers that modulate and transmit the first beam of light and the another of the co-packaged optics modules comprises coherent transceivers that modulate and transmit the second beam of light.

(51) Example 3. The method of examples 1 or 2, wherein the transceivers in one of the co-packaged optics modules and the another of the co-packaged optics modules are both direct detection transceivers.

(52) Example 4. The method of any of examples 1-3, wherein the transceivers in one of the co-packaged optics modules and the another of the co-packaged optics modules are both coherent transceivers.

(53) Example 5. The method of any of examples 1-4, further comprising: receiving, by the co-packaged optics device, a first received beam of light and a second received beam of light; demodulating the first received beam of light using the one of the co-packaged optics modules; and demodulating the second received beam of light using the another of the co-packaged optics modules.

(54) Example 6. The method of any of examples 1-5, wherein the first received beam of light is received by the one of the co-packaged optics modules, and wherein the second received beam of light is received by the another of the co-packaged optics modules.

(55) Example 7. The method of any of examples 1-6, wherein the one or more direct detection transceivers include one or more pulse amplitude modulation (PAM) optical transceivers in a PAM

module of the plurality of co-packaged optics modules.

(56) Example 8. The method of any of examples 1-7, wherein the PAM optical transceivers are configured to modulate and demodulate in different PAM formats.

(57) Example 9. The method of any of examples 1-8, wherein the one or more coherent transceivers include one or more quadrature amplitude modulation (QAM) optical transceivers in a coherent module of the plurality of co-packaged optics modules.

(58) Example 10. The method of any of examples 1-9, wherein the QAM optical transceivers are configured to modulate and demodulate in different QAM formats.

(59) Example 11. The method of any of examples 1-10, wherein the one or more coherent transceivers include one or more phase shift keying (PSK) optical transceivers in a coherent module of the plurality of co-packaged optics modules.

(60) Example 12. The method of any of examples 1-11, wherein the PSK optical transceivers are configured to modulate and demodulate in different PSK formats.

(61) Example 13. The method of any of examples 1-12, wherein the coupler is a spatial mode coupler that separates light based on different optical spatial modes.

(62) Example 14. The method of any of examples 1-13, wherein the coupler is an optical splitter.

(63) Example 15. The method of any of examples 1-14, wherein each of the one or more direct detection transceivers and the one or more coherent transceivers are photonic integrated circuit (PIC) transceivers that are co-packaged with electrical circuitry.

(64) Example 16. The method of any of examples 1-15, wherein the co-packaged optical device comprises an integrated optical-electrical ethernet switch.

(65) Example 17. The method of any of examples 1-16, wherein the integrated optical-electrical ethernet switch and the PIC transceivers are interconnected by an electrical interface.

(66) Example 18. The method of any of examples 1-17, wherein the plurality of beams are generated by one or more external laser assemblies.

(67) Example 19. The method of any of examples 1-18, wherein the one or more external laser assemblies are tunable or fixed light sources.

(68) Example 20. A co-packaged optical device comprising: a plurality of co-packaged optics modules to transmit and receive light in different modulation formats, the plurality of co-packaged optics modules including at least one or more direct detection transceivers or coherent transceivers, the plurality of co-packaged optics modules to receive, from a few mode fiber coupled to the co-packaged optical device, multimode light comprising a plurality of beams in different modes including a first mode of light and a second mode of light; and a coupler to separate the first beam of light from the first mode of the multimode light and separate a second beam of light from the second mode of the multimode light, one of the co-packaged optics modules configured to modulate and transmit the first beam of light and another of the co-packaged optics modules configured to modulate and transmit the second beam of light.

(69) Example 21: A method comprising: receiving, by a co-packaged optical device, multimode light comprising a plurality of beams in different modes including a first mode and a second mode, the multimode light received from a few-mode fiber coupled to the co-packaged optical device, the co-packaged optical device comprising a plurality of co-packaged optics modules configured to transmit and receive light in different modulation formats, the plurality of co-packaged optics modules including at least one or more direct detection transceivers or coherent transceivers; separating, using a few-mode coupler of the co-packaged optical device, a first beam of light from the first mode and a second beam of light from the second mode; modulating the first beam of light using one of the co-packaged optics modules and modulating the second beam of light using another of the co-packaged optics modules; and transmitting the first beam of light using the one of the co-packaged optics modules and transmitting the second beam of light using the another of the co-packaged optics modules.

(70) Example 22. The method of example 21, wherein a transceiver in one of the co-packaged

optics modules is a direct detection transceiver that modulates and transmits the first beam of light, and wherein another transceiver in the another of the co-packaged optics module is a coherent transceiver that modulates and transmits the second beam of light.

(71) Example 23. The method of any of examples 21 or 22, wherein a transceiver in one of the co-packaged optics modules and another transceiver in the another of the co-packaged optics modules are both direct detection transceivers.

(72) Example 24. The method of any of examples 1-23, wherein a transceiver in one of the co-packaged optics modules and another transceiver in the another of the co-packaged optics modules are both coherent transceivers.

(73) Example 25. The method of any of examples 1-24, further comprising: receiving, by the co-packaged optical device, a first received beam of light and a second received beam of light; demodulating the first received beam of light using the one of the co-packaged optics modules; and demodulating the second received beam of light using the another of the co-packaged optics modules.

(74) Example 26. The method of any of examples 1-25, wherein the first received beam of light is received by the one of the co-packaged optics modules, and wherein the second received beam of light is received by the another of the co-packaged optics modules.

(75) Example 27. The method of any of examples 1-26, wherein the one or more direct detection transceivers include one or more pulse amplitude modulation (PAM) optical transceivers in a PAM module of the plurality of co-packaged optics modules.

(76) Example 28. The method of any of examples 1-27, wherein the PAM optical transceivers are configured to modulate and demodulate in different PAM formats.

(77) Example 29. The method of any of examples 1-28, wherein the one or more coherent transceivers include one or more quadrature amplitude modulation (QAM) optical transceivers in a coherent module of the plurality of co-packaged optics modules.

(78) Example 30. The method of any of examples 1-29, wherein the QAM optical transceivers are configured to modulate and demodulate in different QAM formats.

(79) Example 31. The method of any of examples 1-30, wherein the one or more coherent transceivers include one or more phase shift keying (PSK) optical transceivers in a coherent module of the plurality of co-packaged optics modules; and wherein the PSK optical transceivers are configured to modulate and demodulate in different PSK formats.

(80) Example 32. The method of any of examples 1-31, wherein the coupler is a spatial mode coupler that separates light based on different optical spatial modes.

(81) Example 33. The method of any of examples 1-32, wherein the coupler is an optical splitter that splits the light from a single mode fiber to each transceiver.

(82) Example 34. The method of any of examples 1-33, wherein each of the one or more direct detection transceivers and the one or more coherent transceivers are photonic integrated circuit (PIC) transceivers that are co-packaged with electrical circuitry.

(83) Example 35. The method of any of examples 1-34, wherein the co-packaged optical device comprises an integrated optical-electrical ethernet switch.

(84) Example 36. The method of any of examples 1-35, wherein the integrated optical-electrical ethernet switch and the PIC transceivers are interconnected by an electrical interface.

(85) Example 37. The method of any of examples 1-36, wherein the plurality of beams are generated by one or more external laser assemblies.

(86) Example 38. The method of any of examples 1-37, wherein the one or more external laser assemblies are tunable or fixed light sources.

(87) Example 39. The method of any of examples 1-38, wherein the one or more external laser assemblies include one or more of: a distributed feedback laser, a quantum dot laser, an external cavity laser; and wherein the one or more external laser assemblies have different optical characteristics including one or more of linewidth, output power, or relative intensity noise.

(88) Example 40. A co-packaged optical device comprising: a plurality of co-packaged optics modules to transmit and receive single mode light in different modulation formats, the plurality of co-packaged optics modules including at least one or more direct detection transceivers or coherent transceivers, the plurality of co-packaged optics modules to receive, from a few-mode fiber coupled to the co-packaged optical device, multimode light comprising a plurality of beams in different modes including a first mode of light and a second mode of light and the light in each mode can contain multiple wavelength channels; and a coupler to separate the first beam of light from the first mode of the multimode light and separate a second beam of light from the second mode of the multimode light, one of the co-packaged optics modules to modulate and transmit the first beam of light and another of the co-packaged optics modules to modulate and transmit the second beam of light. Other embodiments of this aspect include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods.

(89) In the foregoing detailed description, the method and apparatus of the present inventive subject matter have been described with reference to specific exemplary embodiments thereof. It will, however, be evident that various modifications and changes may be made thereto without departing from the broader spirit and scope of the present inventive subject matter. The present specification and figures are accordingly to be regarded as illustrative rather than restrictive.

Claims

1. A system comprising: multiple light sources configured to generate multiple channels of light; a multiplexer comprising a spatial mode coupler configured to convert light in the multiple channels of light into multiple higher-order spatial modes and to combine the multiple higher-order spatial modes on an optical fiber as multiple respective mode channels; the optical fiber, configured to transmit the multiple channels of light to a sourceless co-packaged optical transceiver; and the sourceless co-packaged optical transceiver, comprising multiple co-packaged optical transceiver modules and a demultiplexer comprising a spatial mode decoupler configured to separate the multiple mode channels based on the respective higher-order spatial modes and to convert light in the multiple mode channels into lower-order spatial modes to be input to the multiple co-packaged optical transceiver modules, wherein the multiple co-packaged optical transceiver modules are configured to modulate the multiple channels of light and to transmit the multiple modulated channels of light to different respective destinations, and wherein a first one of the co-packaged optical transceiver modules utilizes a first modulation format and a second one of the co-packaged optical transceiver modules utilizes a second modulation format that is different from the first modulation format.
2. The system of claim 1, wherein the optical fiber is a few-mode fiber.
3. The system of claim 1, wherein the multiple channels of light comprise first channels of light and second channels of light, and wherein the multiplexer is configured to convert light in the first channels of light into a first higher-order spatial mode corresponding to a first one of the multiple mode channels and to convert light in the second channels of light into a second higher-order spatial mode corresponding to a second one of the multiple mode channels.
4. The system of claim 3, wherein the first channels of light comprise multiple wavelength channels of light having multiple different respective wavelengths.
5. The system of claim 4, wherein the multiple light sources comprise a quantum dot laser to generate the first channels of light, the quantum dot laser being configurable to generate light at any one or more of the multiple different wavelengths.
6. The system of claim 4, wherein the multiple light sources comprise a first set of lasers to generate the first channels of light, and wherein the multiplexer further comprises a single mode coupler configured to combine the first channels of light on a single mode fiber for transmission to

the spatial mode coupler.

7. The system of claim 6, wherein the first set of lasers is configured to generate the multiple wavelength channels of light at multiple respective output powers.

8. The system of claim 6, wherein the first set of lasers comprises a redundancy laser tunable to generate light at any of the multiple wavelengths of the multiple wavelength channels.

9. The system of claim 4, wherein the multiple wavelength channels in the first one of the multiple mode channels are input to the first co-packaged optical transceiver modules, and wherein the first one of the co-packaged optical transceiver modules comprises a wavelength demultiplexer configured to separate the multiple wavelength channels between inputs to multiple respective transceivers of the first one of the co-packaged optical transceiver modules.

10. The system of claim 9, wherein the first modulation format utilized by the first one of the co-packaged optical transceiver modules is a pulse amplitude modulation (PAM) format.

11. The system of claim 10, wherein the multiple transceivers of the first one of the co-packaged optical transceiver modules are of different types configured for PAM data transmission over different respective distances.

12. The system of claim 9, wherein the first modulation format utilized by the first one of the co-packaged optical transceiver modules comprises at least one of phase-shift keying (PSK) or quadrature amplitude modulation (QAM) format.

13. The system of claim 12, wherein the multiple transceivers of the first one of the co-packaged optical transceiver modules are coherent transceivers of different types configured to utilize different respective modulation formats selected among M-ary PSK and M-ary QAM formats.

14. A method comprising: generating, by multiple light sources, multiple channels of light; converting, by a multiplexer, light in the multiple channels of light into multiple higher-order spatial modes; combining, by the multiplexer, the multiple higher-order spatial modes on an optical fiber as multiple respective mode channels; transmitting, by the optical fiber, the multiple channels of light to a source-less co-packaged optical transceiver; separating, by a demultiplexer of the sourceless co-packaged optical transceiver, the multiple mode channels of based on the respective higher-order spatial modes and to convert light in the multiple mode channels into lower-order spatial modes to be input to multiple co-packaged optical transceiver modules of the sourceless co-packaged optical transceiver; modulating, by the multiple co-packaged optical transceiver modules, the multiple channels of light, wherein a first one of the co-packaged optical transceiver modules utilizes a first modulation format and a second one of the co-packaged optical transceiver modules utilizes a second modulation format that is different from the first modulation format; and transmitting, by the multiple co-packaged optical transceiver modules, the multiple modulated channels of light to different respective destinations.

15. The method of claim 14, wherein generating the multiple channels of light comprises generating multiple wavelength channels of light having multiple different respective wavelengths, and wherein the multiple higher-order spatial modes include a first higher-order spatial mode corresponding to a first one of the multiple mode channels.

16. The method of claim 15, wherein the multiple wavelength channels in the first one of the multiple mode channels are input to the first co-packaged optical transceiver modules, the method further comprising: separating the multiple wavelength channels between inputs to multiple respective transceivers of the first one of the co-packaged optical transceiver modules.

17. The method of claim 15, wherein modulating the multiple channels of light and transmitting the multiple modulated channels comprises using multiple transceivers of the first one of the multiple co-packaged optical transceiver modules to modulate the multiple wavelength channels in accordance with a pulse amplitude modulation (PAM) format and transmitting the multiple modulated wavelength channels over multiple different respective distances.

18. The method of claim 15, wherein modulating the multiple channels of light comprises using multiple coherent transceivers of the first one of the multiple co-packaged optical transceiver

modules to modulate the multiple wavelength channels in accordance with multiple respective modulation formats selected among M-ary phase-shift keying (PSK) and M-ary quadrature amplitude modulation (QAM) formats.

19. The method of claim 15, wherein the multiple light sources comprise a quantum dot laser to generate the multiple wavelength channels of light, the quantum dot laser being configurable to generate light at any one or more of the multiple different wavelengths.

20. The method of claim 15, wherein the multiple light sources comprise a first set of lasers to generate the multiple wavelength channels of light, and wherein the multiplexer comprises a spatial mode coupler and a single mode coupler, the single mode coupler being configured to combine the multiple wavelength channels of light on a single mode fiber for transmission to the spatial mode coupler.
